Statistical Analysis on Apartment Rent Prices in Germany

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```r
library(ggplot2)
library(MASS)
library(interactions)
```

```
## Warning: package 'interactions' was built under R version 4.4.3
```

Project 3 Checkpoint

Header and Dimensions of apartments.csv

```r
apartment = read.csv("apartments.csv")
head(apartment)
```

```
##                   regio1 serviceCharge      heatingType telekomTvOffer
## 1         Sachsen_Anhalt            26 district_heating  ONE_YEAR_FREE
## 2                 Hessen           100        heat_pump      ON_DEMAND
## 3            Brandenburg           105      gas_heating  ONE_YEAR_FREE
## 4    Nordrhein_Westfalen            85      gas_heating  ONE_YEAR_FREE
## 5 Mecklenburg_Vorpommern            45  central_heating  ONE_YEAR_FREE
## 6         Sachsen_Anhalt           140  central_heating  ONE_YEAR_FREE
##   newlyConst balcony picturecount pricetrend totalRent yearConstructed
## 1      FALSE    TRUE            7       3.27    282.44            1989
## 2      FALSE   FALSE            4       6.33    700.00            2012
## 3      FALSE   FALSE           10       5.34    489.00            1861
## 4      FALSE    TRUE           11       3.45    390.00            1992
## 5      FALSE   FALSE            6       1.16    294.00            1968
## 6      FALSE    TRUE            6       1.54    560.00            1938
##         firingTypes hasKitchen cellar baseRent livingSpace
## 1  district_heating      FALSE   TRUE   211.44       35.24
## 2       electricity       TRUE  FALSE   600.00       63.00
## 3               gas       TRUE   TRUE   384.00       48.00
## 4 natural_gas_light      FALSE   TRUE   305.00       35.00
## 5  district_heating       TRUE  FALSE   204.00       24.00
## 6               gas      FALSE   TRUE   420.00       70.00
##                           condition  lift baseRentRange   typeOfFlat noRooms
## 1 first_time_use_after_refurbishment FALSE             1 ground_floor       1
## 2                     mint_condition FALSE             4    apartment       2
## 3 first_time_use_after_refurbishment  TRUE             2    apartment       2
## 4                        modernized FALSE             2    apartment       2
## 5                       refurbished  TRUE             1    apartment       1
```

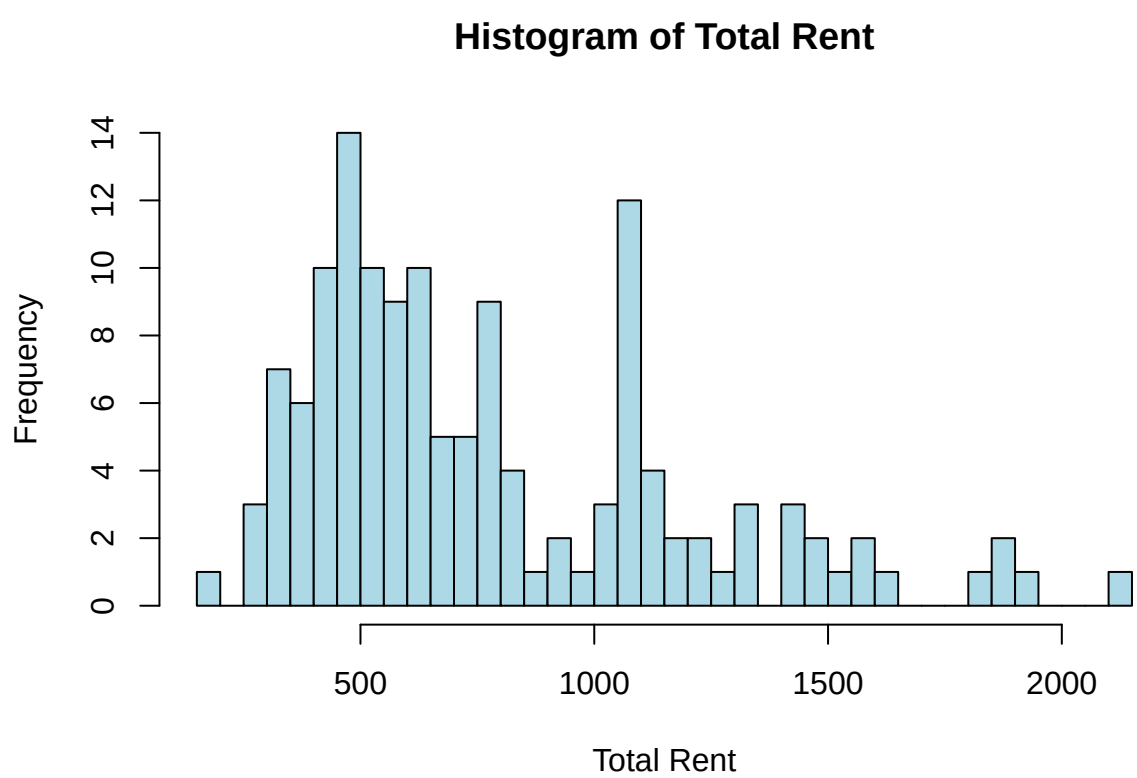
```
## 6                        refurbished FALSE             3    apartment       2
##   floor numberOfFloors noRoomsRange garden livingSpaceRange   date
## 1     2              3            1  FALSE                1 19-May
## 2     2              3            2  FALSE                3 18-Sep
## 3     1              3            2   TRUE                2 20-Feb
## 4     2              2            2  FALSE                1 19-Oct
## 5     8             11            1  FALSE                1 19-May
## 6     2              2            2  FALSE                3 19-May
```

```r
dim(apartment)
```

```
## [1] 138  26
```

Histogram:

```r
hist(apartment$totalRent,
     main = "Histogram of Total Rent",
     xlab = "Total Rent",
     col = "lightblue",
     breaks = 30)
```



The histogram seems to show the relationship between Total Rent and the Frequency of each rent bin as having a right skewed distribution and having a cluster near the 500 euros rent range, with most rent values falling below 1000 euros. There are possible outliers, notably in the 2000 euros and above range.

```r
summary(apartment$totalRent)
```

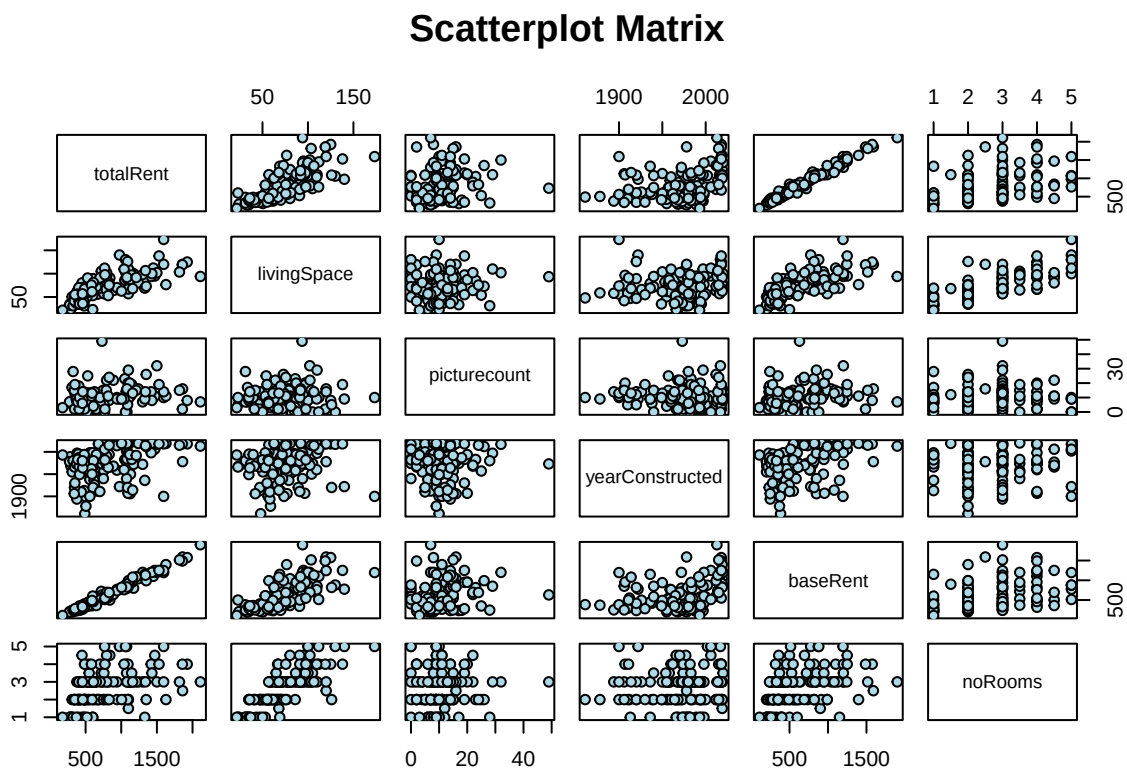
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   177.2   480.0   647.5   784.9  1075.0  2110.0
```

```r
sd(apartment$totalRent, na.rm = TRUE)
```

```
## [1] 405.4092
```

```r
scatterMatrix = apartment[, c("totalRent", "livingSpace", "picturecount", "yearConstructed", "baseRent"

pairs(scatterMatrix,
      main = "Scatterplot Matrix",
      pch = 21,
      bg = "lightblue")
```



Scatterplot Interpretation: totalRent and livingSpace seems to have a strong positive linear relationship. totalRent and baseRent also seems to have a strong positive linear relationship. totalRent and noRooms seems to have a somewhat positive discrete relationship due to noRooms being a discrete variable. totalRent and yearConstructed possibly has a positive linear relationship, though it is hard to tell since it is rather weak. totalRent and picturecount seems to have none to very weak positive relationship. There seems to be a few predictors, namely livingSpace and baseRent that could help provide a linear model for the data.

Linear Model:

```r
apartment$condition = as.factor(apartment$condition)
model1 = lm(totalRent ~ livingSpace + noRooms + condition + hasKitchen + yearConstructed, data = apartme

summary(model1)
```

```
##
## Call:
## lm(formula = totalRent ~ livingSpace + noRooms + condition +
##     hasKitchen + yearConstructed, data = apartment)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -499.13 -104.08  -17.34   88.21  733.71
##
## Coefficients:
##                                            Estimate Std. Error t value
## (Intercept)                              -1498.2390  1132.4084  -1.323
## livingSpace                                 12.5906     1.0056  12.521
## noRooms                                    -53.8526    26.4922  -2.033
## conditionfirst_time_use_after_refurbishment  -254.4959    83.6616  -3.042
## conditionfully_renovated                   -289.6457    79.9872  -3.621
## conditionmint_condition                    -109.8371    73.0353  -1.504
## conditionmodernized                        -268.1170    82.2263  -3.261
## conditionneed_of_renovation                -421.4245   150.4019  -2.802
## conditionnegotiable                        -590.7707   198.8221  -2.971
## conditionrefurbished                       -386.4065    86.4972  -4.467
## conditionwell_kept                         -376.0137    71.0617  -5.291
## hasKitchenTRUE                              238.5575    35.6867   6.685
## yearConstructed                               0.8564     0.5585   1.533
##                                          Pr(>|t|)
## (Intercept)                              0.188232
## livingSpace                               < 2e-16 ***
## noRooms                                  0.044193 *
## conditionfirst_time_use_after_refurbishment 0.002865 **
## conditionfully_renovated                 0.000425 ***
## conditionmint_condition                  0.135132
## conditionmodernized                      0.001432 **
## conditionneed_of_renovation              0.005888 **
## conditionnegotiable                      0.003557 **
## conditionrefurbished                     1.75e-05 ***
## conditionwell_kept                       5.26e-07 ***
## hasKitchenTRUE                           6.87e-10 ***
## yearConstructed                          0.127715
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 185.1 on 125 degrees of freedom
## Multiple R-squared:  0.8097, Adjusted R-squared:  0.7915
## F-statistic: 44.33 on 12 and 125 DF,  p-value: < 2.2e-16
```

```r
levels(apartment$condition)
```

```
## [1] "first_time_use"                      "first_time_use_after_refurbishment"
```

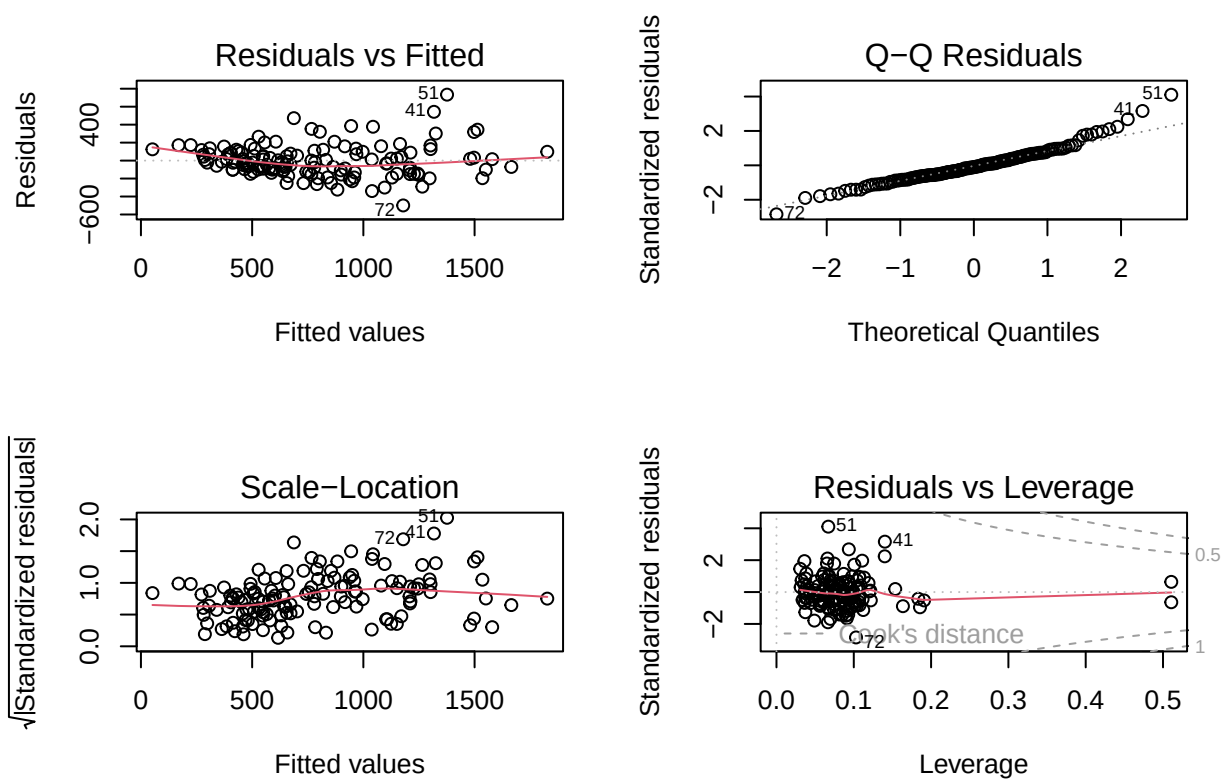
```
## [3] "fully_renovated"                    "mint_condition"
## [5] "modernized"                         "need_of_renovation"
## [7] "negotiable"                         "refurbished"
## [9] "well_kept"
```

The initial model I used is predicted totalRent = beta0 + beta1* livingSpace + beta2* noRooms + beta3* condition + beta4* hasKitchen + beta5*yearConstructed. There is an adjusted R-squared value of 0.7915, meaning about 79.15% of the variability can be explained by the model. I used the adjusted value because it is possible still that some of the predictors do not improve the model. One quantitative predictor is livingSpace. Based on the model summary, for every additional square meter of livingSpace, the total rent on average increases by approximately 12.5906 euros, holding all other variables constant. One categorical predictor is the condition of the apartment. The* baseline variable* is first_time_use, so all of the "condition" predictors are based off of the baseline. For example, the coefficient for conditionrefurbished means that the apartments that are considered refurbished are associated with a -386.4065 euro decrease in rent on average compared to first time use apartments, holding other variables constant.

Diagnostic Plots

```r
par(mfrow = c(2, 2))
plot(model1)
```

```
## Warning: not plotting observations with leverage one:
##   121
```

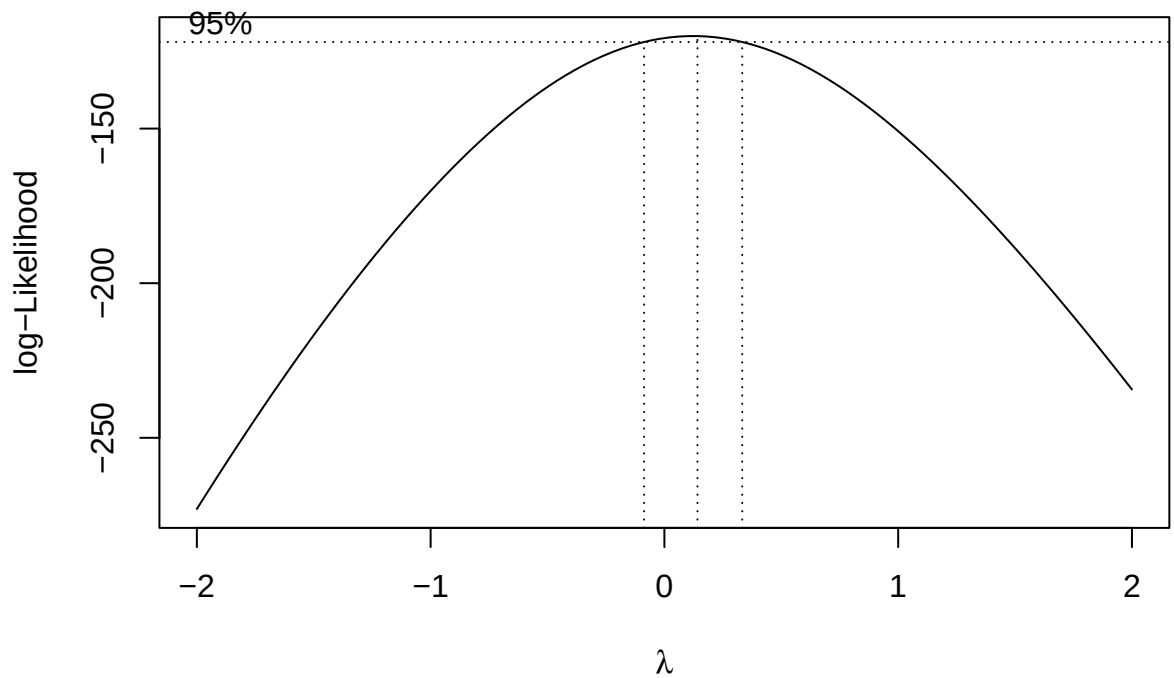


The residuals vs fitted graph shows most dots randomly scattered along the horizontal line at 0. However, there is a slight curve around the points 51, 41, and 72, which shows that there is slight heteroscedasticity.

The q-q residuals seem approximately normal, with a slight skew on the right hand side tail end. The scale Locaiton plot shows most points being relatively horizontal with a fairly constant spread across the fitted values. the points 51, 41, 72 might be somewhat outliers, but there is little to no heteroscedasticity. The residuals vs leverage plot shows most points clustered on the left, signalling low leverage for most points. There are two points that are above 0.5 in leverage, meaning they are high leverage points. Points 51 and 41 as well as a third unamed point are above the |2| residual mark, but none of the points exceed the cook's distance of 1, so none of the points seem to need removal. The data does not seem to require removal of any points as of now.

Box Cox

```
boxcox(model1, lambda = seq(-2, 2, 0.1))
```




```
bc = boxcox(model1, lambda = seq(-2, 2, 0.1))
lambda_best = bc$x[which.max(bc$y)]
lambda_best
```

```
## [1] 0.1414141
```

Given the above box-cox plot, it seems reasonable to not perform any transformations, since the optimal lambda is 0.14141. No transformation would indicate a lamba of 0, which is well within the 95% confidence interval depicted above, and is closer than a square root transformation, which would have a lambda of 0.5.

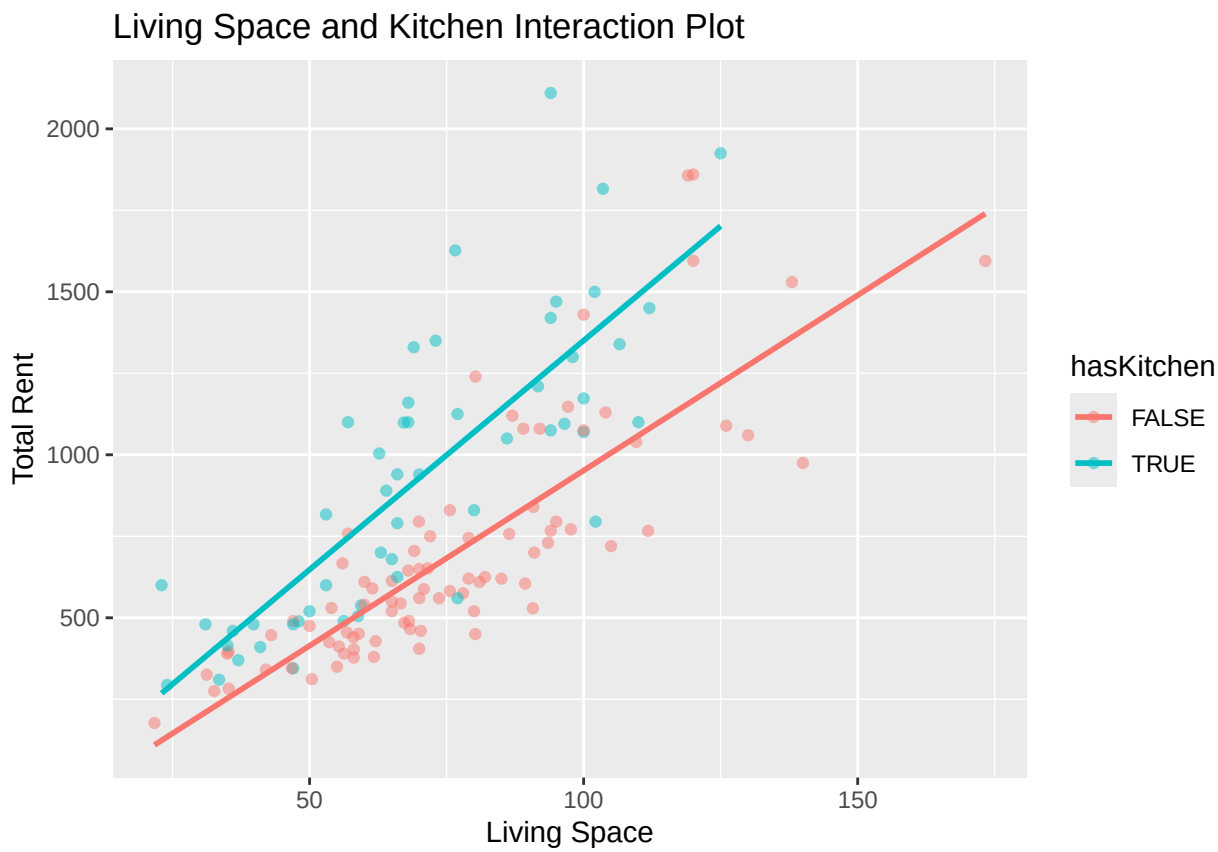
Interaction Term

```r
model2 = lm(totalRent ~ livingSpace * hasKitchen + noRooms + condition + yearConstructed, data = apartme
summary(model2)
```

```
##
## Call:
## lm(formula = totalRent ~ livingSpace * hasKitchen + noRooms +
##     condition + yearConstructed, data = apartment)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -493.72  -99.64  -20.79   82.22  701.83
##
## Coefficients:
##                                            Estimate Std. Error t value
## (Intercept)                               -935.2973  1148.1272  -0.815
## livingSpace                                 11.7838     1.0625  11.091
## hasKitchenTRUE                              37.7721   101.1212   0.374
## noRooms                                    -59.1727    26.2507  -2.254
## conditionfirst_time_use_after_refurbishment -270.5289    82.8653  -3.265
## conditionfully_renovated                   -314.5037    79.7627  -3.943
## conditionmint_condition                    -124.8358    72.3848  -1.725
## conditionmodernized                        -279.0645    81.2676  -3.434
## conditionneed_of_renovation                -435.4221   148.4946  -2.932
## conditionnegotiable                        -556.4430   196.7749  -2.828
## conditionrefurbished                       -394.3052    85.3972  -4.617
## conditionwell_kept                         -391.0527    70.4498  -5.551
## yearConstructed                               0.6168     0.5623   1.097
## livingSpace:hasKitchenTRUE                    2.7521     1.2993   2.118
##                                           Pr(>|t|)
## (Intercept)                               0.416847
## livingSpace                                < 2e-16 ***
## hasKitchenTRUE                            0.709390
## noRooms                                   0.025943 *
## conditionfirst_time_use_after_refurbishment 0.001417 **
## conditionfully_renovated                  0.000134 ***
## conditionmint_condition                   0.087089 .
## conditionmodernized                       0.000810 ***
## conditionneed_of_renovation               0.004009 **
## conditionnegotiable                       0.005467 **
## conditionrefurbished                      9.57e-06 ***
## conditionwell_kept                        1.65e-07 ***
## yearConstructed                           0.274819
## livingSpace:hasKitchenTRUE                0.036167 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 182.6 on 124 degrees of freedom
## Multiple R-squared:  0.8164, Adjusted R-squared:  0.7971
## F-statistic:  42.4 on 13 and 124 DF,  p-value: < 2.2e-16
```

```r
interaction_plot = ggplot(apartment, aes(x = livingSpace, y = totalRent, color = hasKitchen)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
```

```
  labs(title = "Living Space and Kitchen Interaction Plot", x = "Living Space", y = "Total Rent")
interaction_plot
```



Given the interpretation plot, we can conclude that total rent increases as living space increases, regardless of if it has a kitchen or not. It does seem that the two slopes are incredidbly close to each other, so we will have to see whether or not we should include the interaction term using further tests.

Model Selection and AIC

```
model_null = lm(totalRent ~ 1, data = apartment)

model_full = lm(totalRent ~ livingSpace * hasKitchen + noRooms + condition + yearConstructed, data = apa

model_forward = step(model_null,
                     scope = list(lower = model_null, upper = model_full),
                     direction = "forward")
```

```
## Start:  AIC=1658.35
## totalRent ~ 1
##
##                    Df Sum of Sq      RSS    AIC
## + livingSpace       1  12214938 10301924 1552.4
## + condition         8   7132455 15384406 1621.8
## + noRooms           1   4036477 18480384 1633.1
## + yearConstructed   1   2582250 19934611 1643.5
## + hasKitchen        1   1841793 20675068 1648.6
```

```
## <none>                          22516861 1658.3
##
## Step:  AIC=1552.44
## totalRent ~ livingSpace
##
##                   Df Sum of Sq      RSS    AIC
## + condition        8   4148447  6153476 1497.3
## + hasKitchen       1   3040406  7261517 1506.2
## + yearConstructed  1   1338594  8963329 1535.2
## + noRooms          1   1136524  9165400 1538.3
## <none>                          10301924 1552.4
##
## Step:  AIC=1497.33
## totalRent ~ livingSpace + condition
##
##                   Df Sum of Sq     RSS    AIC
## + hasKitchen       1   1648072 4505404 1456.3
## + noRooms          1    303460 5850016 1492.3
## <none>                          6153476 1497.3
## + yearConstructed  1     30325 6123151 1498.7
##
## Step:  AIC=1456.31
## totalRent ~ livingSpace + condition + hasKitchen
##
##                           Df Sum of Sq     RSS    AIC
## + livingSpace:hasKitchen  1    158732 4346672 1453.4
## + noRooms                 1    140308 4365096 1453.9
## + yearConstructed         1     79263 4426141 1455.9
## <none>                                4505404 1456.3
##
## Step:  AIC=1453.36
## totalRent ~ livingSpace + condition + hasKitchen + livingSpace:hasKitchen
##
##                   Df Sum of Sq     RSS    AIC
## + noRooms          1    171640 4175032 1449.8
## <none>                          4346672 1453.4
## + yearConstructed  1     42325 4304347 1454.0
##
## Step:  AIC=1449.8
## totalRent ~ livingSpace + condition + hasKitchen + noRooms +
##     livingSpace:hasKitchen
##
##                   Df Sum of Sq     RSS    AIC
## <none>                          4175032 1449.8
## + yearConstructed  1     40120 4134911 1450.5
```

```r
summary(model_forward)
```

```
##
## Call:
## lm(formula = totalRent ~ livingSpace + condition + hasKitchen +
##     noRooms + livingSpace:hasKitchen, data = apartment)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -492.27 -112.69  -20.43   88.18  707.42
##
## Coefficients:
##                                          Estimate Std. Error t value
## (Intercept)                               320.151     90.484   3.538
## livingSpace                                11.691      1.060  11.029
## conditionfirst_time_use_after_refurbishment -306.520     76.154  -4.025
## conditionfully_renovated                  -350.484     72.765  -4.817
## conditionmint_condition                   -137.302     71.545  -1.919
## conditionmodernized                       -308.485     76.776  -4.018
## conditionneed_of_renovation               -470.031    145.222  -3.237
## conditionnegotiable                       -562.273    196.863  -2.856
## conditionrefurbished                      -437.385     75.891  -5.763
## conditionwell_kept                        -425.110     63.290  -6.717
## hasKitchenTRUE                             13.789     98.809   0.140
## noRooms                                   -59.551     26.270  -2.267
## livingSpace:hasKitchenTRUE                  3.039      1.274   2.385
##                                          Pr(>|t|)
## (Intercept)                              0.000566 ***
## livingSpace                               < 2e-16 ***
## conditionfirst_time_use_after_refurbishment 9.81e-05 ***
## conditionfully_renovated                 4.14e-06 ***
## conditionmint_condition                  0.057251 .
## conditionmodernized                      0.000101 ***
## conditionneed_of_renovation              0.001548 **
## conditionnegotiable                      0.005023 **
## conditionrefurbished                     6.07e-08 ***
## conditionwell_kept                       5.85e-10 ***
## hasKitchenTRUE                           0.889238
## noRooms                                  0.025114 *
## livingSpace:hasKitchenTRUE               0.018559 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 182.8 on 125 degrees of freedom
## Multiple R-squared:  0.8146, Adjusted R-squared:  0.7968
## F-statistic: 45.76 on 12 and 125 DF,  p-value: < 2.2e-16
```

I used AIC and forward searching as my model selection method and metric. This is because there are a lot of variables ranging from categorical to quantitative, and this helps reduce overfitting by adding a penalty when there are too many variables.

Model Comparison

```r
summary(model_forward)
```

```
##
## Call:
## lm(formula = totalRent ~ livingSpace + condition + hasKitchen +
##     noRooms + livingSpace:hasKitchen, data = apartment)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -492.27 -112.69  -20.43   88.18  707.42
##
## Coefficients:
##                                          Estimate Std. Error t value
## (Intercept)                               320.151     90.484   3.538
## livingSpace                                11.691      1.060  11.029
## conditionfirst_time_use_after_refurbishment -306.520     76.154  -4.025
## conditionfully_renovated                  -350.484     72.765  -4.817
## conditionmint_condition                   -137.302     71.545  -1.919
## conditionmodernized                       -308.485     76.776  -4.018
## conditionneed_of_renovation               -470.031    145.222  -3.237
## conditionnegotiable                       -562.273    196.863  -2.856
## conditionrefurbished                      -437.385     75.891  -5.763
## conditionwell_kept                        -425.110     63.290  -6.717
## hasKitchenTRUE                             13.789     98.809   0.140
## noRooms                                   -59.551     26.270  -2.267
## livingSpace:hasKitchenTRUE                  3.039      1.274   2.385
##                                          Pr(>|t|)
## (Intercept)                              0.000566 ***
## livingSpace                               < 2e-16 ***
## conditionfirst_time_use_after_refurbishment 9.81e-05 ***
## conditionfully_renovated                 4.14e-06 ***
## conditionmint_condition                  0.057251 .
## conditionmodernized                      0.000101 ***
## conditionneed_of_renovation              0.001548 **
## conditionnegotiable                      0.005023 **
## conditionrefurbished                     6.07e-08 ***
## conditionwell_kept                       5.85e-10 ***
## hasKitchenTRUE                           0.889238
## noRooms                                  0.025114 *
## livingSpace:hasKitchenTRUE               0.018559 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 182.8 on 125 degrees of freedom
## Multiple R-squared:  0.8146, Adjusted R-squared:  0.7968
## F-statistic: 45.76 on 12 and 125 DF,  p-value: < 2.2e-16
```

```r
summary(model1)
```

```
##
## Call:
## lm(formula = totalRent ~ livingSpace + noRooms + condition +
##     hasKitchen + yearConstructed, data = apartment)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -499.13 -104.08  -17.34   88.21  733.71
##
## Coefficients:
##                                            Estimate Std. Error t value
## (Intercept)                              -1498.2390  1132.4084  -1.323
## livingSpace                                 12.5906     1.0056  12.521
## noRooms                                    -53.8526    26.4922  -2.033
```

```
## conditionfirst_time_use_after_refurbishment  -254.4959    83.6616  -3.042
## conditionfully_renovated                     -289.6457    79.9872  -3.621
## conditionmint_condition                      -109.8371    73.0353  -1.504
## conditionmodernized                          -268.1170    82.2263  -3.261
## conditionneed_of_renovation                  -421.4245   150.4019  -2.802
## conditionnegotiable                          -590.7707   198.8221  -2.971
## conditionrefurbished                         -386.4065    86.4972  -4.467
## conditionwell_kept                           -376.0137    71.0617  -5.291
## hasKitchenTRUE                                238.5575    35.6867   6.685
## yearConstructed                                 0.8564     0.5585   1.533
##                                            Pr(>|t|)
## (Intercept)                                0.188232
## livingSpace                                 < 2e-16 ***
## noRooms                                    0.044193 *
## conditionfirst_time_use_after_refurbishment 0.002865 **
## conditionfully_renovated                   0.000425 ***
## conditionmint_condition                    0.135132
## conditionmodernized                        0.001432 **
## conditionneed_of_renovation                0.005888 **
## conditionnegotiable                        0.003557 **
## conditionrefurbished                       1.75e-05 ***
## conditionwell_kept                         5.26e-07 ***
## hasKitchenTRUE                             6.87e-10 ***
## yearConstructed                            0.127715
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 185.1 on 125 degrees of freedom
## Multiple R-squared:  0.8097, Adjusted R-squared:  0.7915
## F-statistic: 44.33 on 12 and 125 DF,  p-value: < 2.2e-16
```

Here, we can see that the new model selected through the model selection method and metric while also having the interaction term has an adjusted r^2 value of 0.7968, while the original linear model has an r^2 value of 0.7915. The new model has a slightly higher r^2 value, though it seems both models are valid representations of the data.

Model Analysis

Why do we select this model?

I selected the model totalRent ~ livingSpace + condition + hasKitchen + noRooms , since it has a very similar AIC value as well as nearly the same r^2 value as the model with the interaction term, meaning that this model balances complexity with predictive power. What really was the deciding factor was the VIF further in this report, where the interaction term had a large collinearity with the model, so that's why I decided to go with the original model.

Fitted Model predicted totalRent = -1498.24 + 12.59 * livingSpace - 53.85 * noRooms + 238.56 * hasKitchen + 0.86 * yearConstructed - 254.50*conditionfirst_time_use_after_refurbishment -289.65* conditionfully_renovated -109.84 * conditionmint_condition -268.12 * condition modernized -421.42 * conditionneed_of_renovation -590.77 * conditionnegotiable -386.41 * conditionrefurbished -376.01 * condition well_kept

N and P The n for the model is 138 total observations, while the p is 13 parameters.

Model Complexity and Size We have about 10.6 observations per parameter, which seems to be a reasonable size. (138/13) Typically, if the observations per parameter lies between 10-20, the complexity and size can be reasonable.

Standard Deviations and y residuals

```
sd(apartment$totalRent, na.rm = TRUE)
```

```
## [1] 405.4092
```

```
final_model = model1
summary(final_model)$sigma
```

```
## [1] 185.1379
```

Since the residual standard deviation is much smaller than the raw standard deviation, the model seems to explain much of the variance in the model, and this aligns with the r^2 value.

VIF

```
apartment$hasKitchen = as.numeric(apartment$hasKitchen)

# VIF for livingSpace
vif_livingSpace = summary(lm(livingSpace ~ hasKitchen + noRooms, data = apartment))$r.squared
vif_livingSpace = 1 / (1 - vif_livingSpace)

# VIF for hasKitchen
vif_hasKitchen = summary(lm(hasKitchen ~ livingSpace + noRooms, data = apartment))$r.squared
vif_hasKitchen = 1 / (1 - vif_hasKitchen)

# VIF for noRooms
vif_noRooms = summary(lm(noRooms ~ livingSpace + hasKitchen, data = apartment))$r.squared
vif_noRooms = 1 / (1 - vif_noRooms)

vif_values = c(
  livingSpace = vif_livingSpace,
  hasKitchen = vif_hasKitchen,
  noRooms = vif_noRooms
)

print(vif_values)
```

```
## livingSpace  hasKitchen     noRooms
##    2.466857    1.048151    2.541121
```

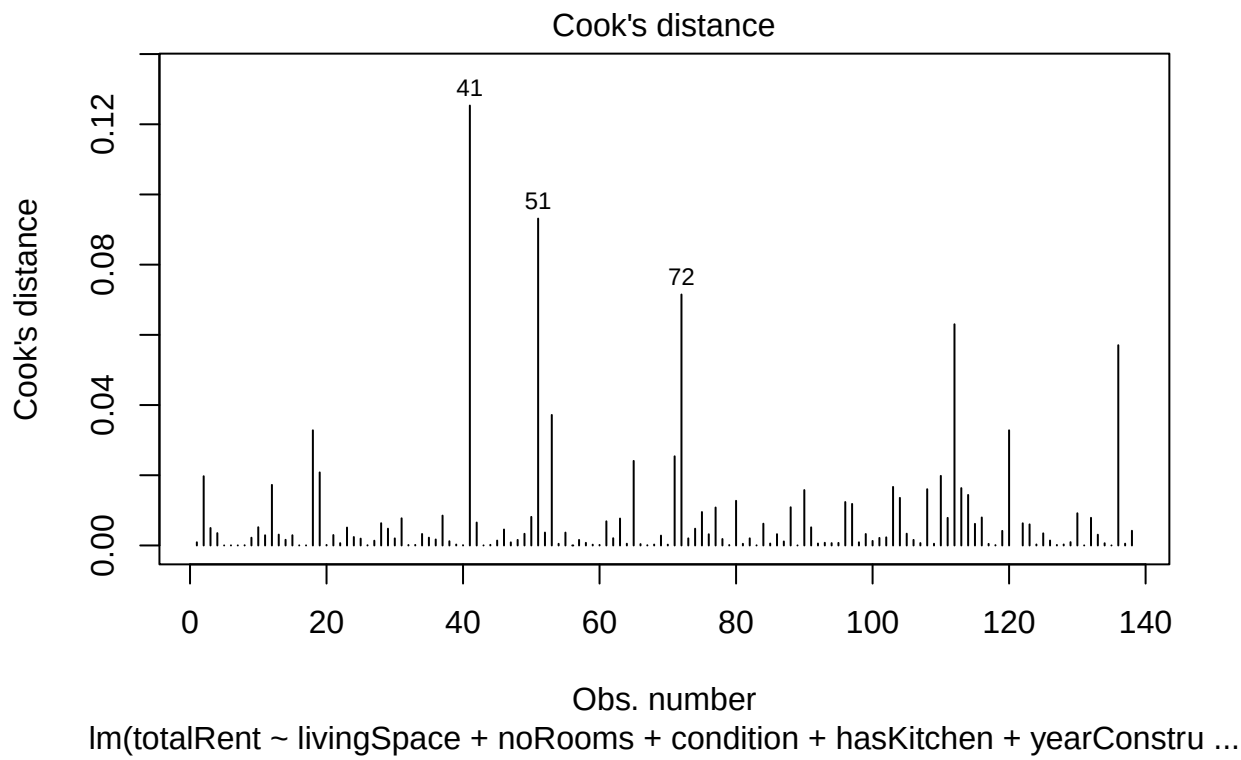
After removing the interaction term, the vif of all the predictors lie below 5, which indicates low collinearity. This aligns with the model that I chose.

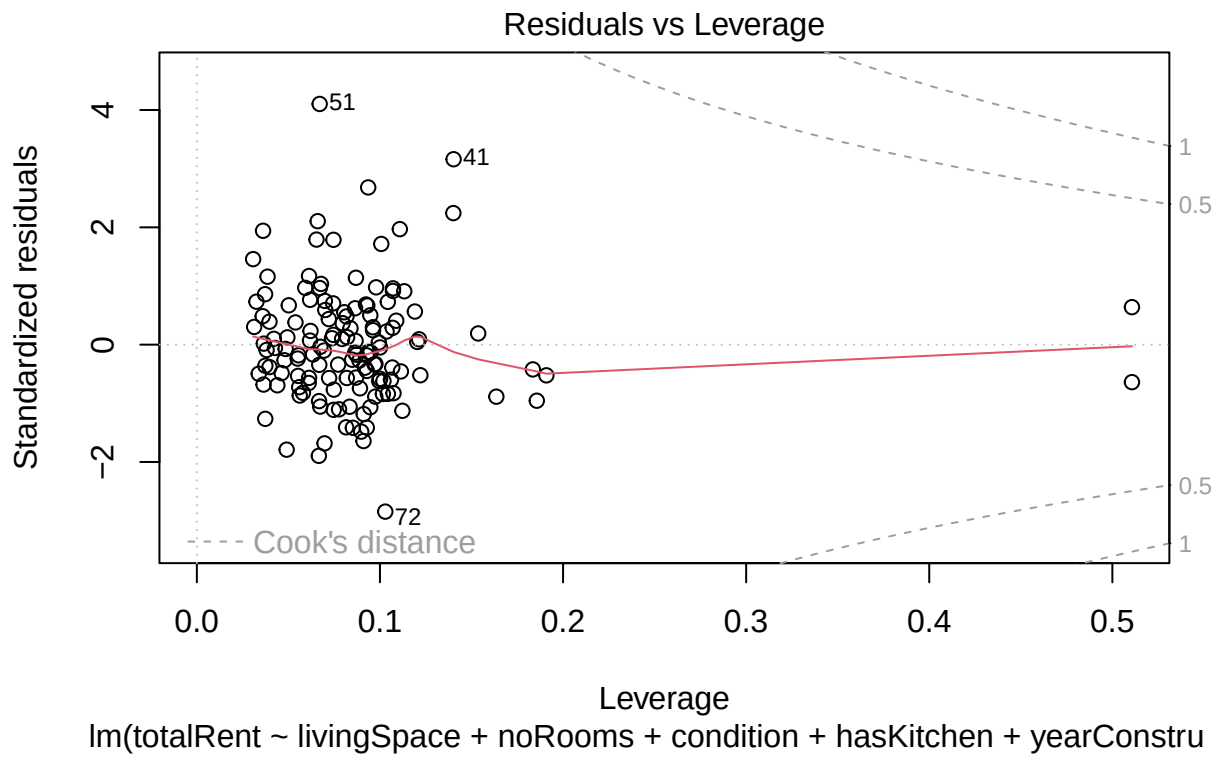
Model Assumptions As shown from the diagnostic plots from the first half of the report, it seems that the model assumptions are met, as none of the diagnostic plots show enough evidence of any problem with the 4 main assumptions.

Influence

```r
plot(model1, which = 4:5)
```

```
## Warning: not plotting observations with leverage one:
##   121
```





There are a few points that have large influence, such as points 153, 53, and 148. However, I will choose to keep them, knowing the context of the apartments such as the possibility of having large luxury apartments or other scenarios that can be justifiably kept, and given that they lie within the Cook's line. There is one point with high leverage, 121, that was excluded by the model.

RSME Calculation

```
model = lm(totalRent ~ livingSpace + hasKitchen + noRooms, data = apartment)
n = nrow(apartment)
squared_errors = numeric(n)

for (i in 1:n) {
  train_data = apartment[-i, ]
  test_data = apartment[i, ]
  loocv_model = lm(totalRent ~ livingSpace + hasKitchen + noRooms, data = train_data)
  prediction = predict(loocv_model, newdata = test_data)
  squared_errors[i] = (test_data$totalRent - prediction)^2
}

loocv_rmse = sqrt(mean(squared_errors))
print(paste("LOOCV RMSE:", round(loocv_rmse, 2)))
```

```
## [1] "LOOCV RMSE: 228.57"
```

The RMSE is about 228.57 euros, which is roughly 15% of the intercept, which is acceptable for our model given the number of data points and predictors.

T-Test For the statistical test, I will perform a t-test on the variable noRooms to see if it affects rent

```r
summary(model1)
```

```
##
## Call:
## lm(formula = totalRent ~ livingSpace + noRooms + condition +
##     hasKitchen + yearConstructed, data = apartment)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -499.13 -104.08  -17.34   88.21  733.71
##
## Coefficients:
##                                            Estimate Std. Error t value
## (Intercept)                              -1498.2390  1132.4084  -1.323
## livingSpace                                 12.5906     1.0056  12.521
## noRooms                                    -53.8526    26.4922  -2.033
## conditionfirst_time_use_after_refurbishment  -254.4959    83.6616  -3.042
## conditionfully_renovated                   -289.6457    79.9872  -3.621
## conditionmint_condition                    -109.8371    73.0353  -1.504
## conditionmodernized                        -268.1170    82.2263  -3.261
## conditionneed_of_renovation                -421.4245   150.4019  -2.802
## conditionnegotiable                        -590.7707   198.8221  -2.971
## conditionrefurbished                       -386.4065    86.4972  -4.467
## conditionwell_kept                         -376.0137    71.0617  -5.291
## hasKitchenTRUE                              238.5575    35.6867   6.685
## yearConstructed                               0.8564     0.5585   1.533
##                                          Pr(>|t|)
## (Intercept)                              0.188232
## livingSpace                               < 2e-16 ***
## noRooms                                  0.044193 *
## conditionfirst_time_use_after_refurbishment 0.002865 **
## conditionfully_renovated                 0.000425 ***
## conditionmint_condition                  0.135132
## conditionmodernized                      0.001432 **
## conditionneed_of_renovation              0.005888 **
## conditionnegotiable                      0.003557 **
## conditionrefurbished                     1.75e-05 ***
## conditionwell_kept                       5.26e-07 ***
## hasKitchenTRUE                           6.87e-10 ***
## yearConstructed                          0.127715
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 185.1 on 125 degrees of freedom
## Multiple R-squared:  0.8097, Adjusted R-squared:  0.7915
## F-statistic: 44.33 on 12 and 125 DF,  p-value: < 2.2e-16
```

Null Hypothesis: The number of rooms does not have a statistically significant impact on the predicted average cost of the total rent, holding all other variables constant. Alternative Hypothesis: The number of rooms does have a statistically significant impact on the predicted average cost of the total rent, holding all other variables constant. Since noRooms has a t value of -2.033 and a p-value of 0.044, this is less than the alpha level of 0.05, which means that we will reject the null hypothesis, and can conclude that the number of rooms does in fact have a significant impact on improving our model.